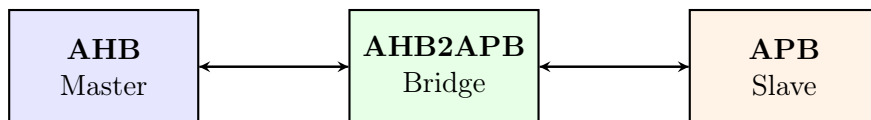


AHB2APB Bridge Specification

Version 1.0

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This document specifies the functional requirements and design of the AHB2APB Bridge.
For verification requirements, refer to the VEGA-generated test plan.

Revision History

Version	Date	Author	Description
1.0	Feb 2026	Vikash Kumar	Initial release

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1 Introduction

1.1 Purpose

This document specifies the functional requirements, interface signals, register definitions, and operational behavior of the AHB2APB Bridge. The bridge provides protocol conversion between the AMBA Advanced High-performance Bus (AHB) and the AMBA Advanced Peripheral Bus (APB).

The specification serves as the authoritative reference for:

- RTL implementation
- Verification test plan generation (via VEGA framework)
- Integration guidelines
- Student learning projects

1.2 Scope

This specification covers:

- Single AHB master to single APB slave configuration
- 32-bit address and data widths
- Single and burst transfer support (INCR, WRAP)
- Sequential burst-to-APB conversion
- Configuration and status registers
- Error detection and reporting

This specification does not cover:

- Multiple APB slave support (future enhancement)
- Clock domain crossing (assumes single clock domain)
- Parity or ECC protection

1.3 Definitions and Acronyms

Term	Definition
AHB	Advanced High-performance Bus
APB	Advanced Peripheral Bus
AMBA	Advanced Microcontroller Bus Architecture
INCR	Incrementing burst
WRAP	Wrapping burst
NONSEQ	Non-sequential transfer (first beat)
SEQ	Sequential transfer (burst continuation)
DUT	Device Under Test
FSM	Finite State Machine

1.4 References

1. ARM AMBA 3 AHB-Lite Protocol Specification (ARM IHI 0033)
2. ARM AMBA 3 APB Protocol Specification (ARM IHI 0024)
3. VEGA Framework User Guide

2 Functional Overview

2.1 Block Diagram

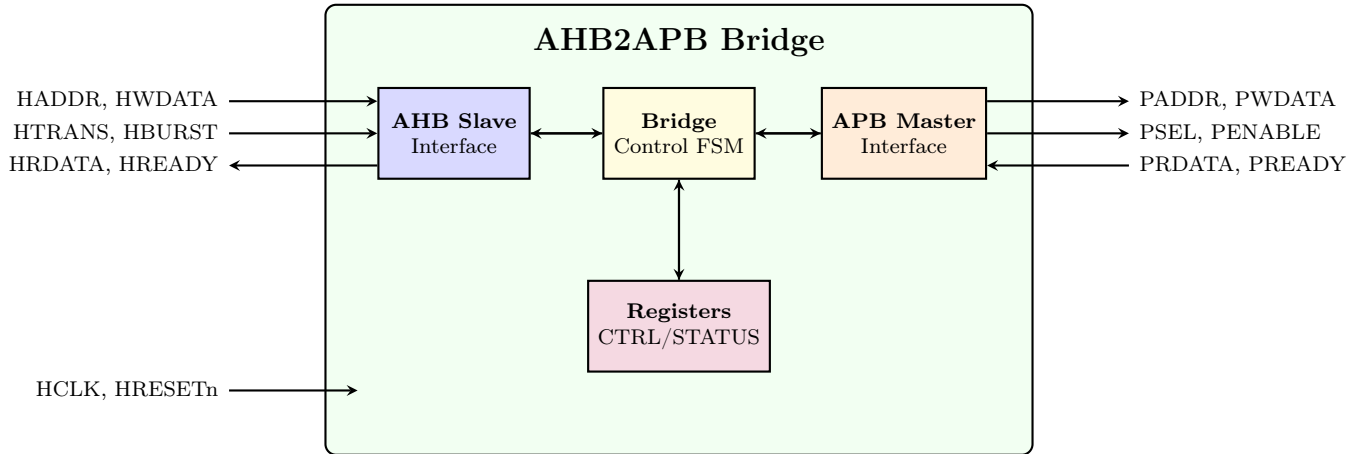


Figure 1: AHB2APB Bridge Block Diagram

2.2 Features Summary

Feature	Description
Protocol Conversion	AHB-Lite to APB protocol translation
Address Width	32-bit
Data Width	32-bit (AHB and APB)
Burst Support	SINGLE, INCR, INCR4, INCR8, INCR16, WRAP4, WRAP8, WRAP16
Burst Handling	Sequential (one APB transfer per AHB beat)
Wait States	APB PREADY propagates to AHB HREADY
Error Handling	Address, Timeout, Protocol errors
Registers	CTRL, STATUS, ERROR_ADDR, ERROR_INFO

Table 1: Feature Summary

2.3 Limitations

- Single APB slave only (no address decoding for multiple slaves)
- Same clock domain for AHB and APB ($HCLK = PCLK$)
- No parity or ECC protection
- Maximum burst length: 16 beats
- No exclusive access support

3 Interface Signals

3.1 AHB Slave Interface

The bridge presents an AHB slave interface to the AHB master.

Signal	Width	Dir	Description
HCLK	1	In	Bus clock. All signals sampled on rising edge.
HRESETn	1	In	Active-low reset. Async assert, sync deassert.
HADDR	32	In	Byte address of transfer.
HTRANS	2	In	Transfer type: IDLE(00), BUSY(01), NONSEQ(10), SEQ(11).
HWRITE	1	In	Transfer direction: 0=Read, 1=Write.
HSIZE	3	In	Transfer size: 000=Byte, 001=Halfword, 010=Word.
HBURST	3	In	Burst type (see Table 3).
HWDATA	32	In	Write data bus.
HSEL	1	In	Slave select. Must be HIGH for bridge to respond.
HREADY_IN	1	In	Previous transfer complete. Used for pipeline.
HRDATA	32	Out	Read data bus.
HREADY_OUT	1	Out	Transfer complete. LOW inserts wait states.
HRESP	1	Out	Transfer response: 0=OKAY, 1=ERROR.

Table 2: AHB Slave Interface Signals

HBURST	Type	Beats
000	SINGLE	1
001	INCR	Undefined
010	WRAP4	4
011	INCR4	4
100	WRAP8	8
101	INCR8	8
110	WRAP16	16
111	INCR16	16

Table 3: HBURST Encoding

3.2 APB Master Interface

The bridge presents an APB master interface to the APB slave.

Signal	Width	Dir	Description
PCLK	1	In	APB clock (same as HCLK in this design).
PRESETn	1	In	Active-low reset (same as HRESETn).
PADDR	32	Out	Byte address of transfer.
PSEL	1	Out	Slave select. HIGH during SETUP and ACCESS.
PENABLE	1	Out	Enable. LOW in SETUP, HIGH in ACCESS phase.
PWRITE	1	Out	Transfer direction: 0=Read, 1=Write.
PWDATA	32	Out	Write data bus.
PRDATA	32	In	Read data bus.
PREADY	1	In	Slave ready. LOW inserts wait states.
PSLVERR	1	In	Slave error. HIGH indicates transfer error.

Table 4: APB Master Interface Signals

3.3 Clock and Reset

- **Clock Domain:** Single clock domain. HCLK and PCLK are the same clock.
- **Reset:** Active-low, asynchronous assert, synchronous deassert.
- **Reset State:** All outputs driven to safe state (see Section 6).

4 Register Interface

The bridge includes configuration and status registers accessible via the AHB interface. Registers are accessed when HADDR[31:12] matches the bridge base address and HADDR[11:8] = 0xF (register space).

4.1 Register Map

Offset	Name	Access	Description
0xF00	CTRL	R/W	Control register
0xF04	STATUS	R/W1C	Status register
0xF08	ERROR_ADDR	RO	Error address capture
0xF0C	ERROR_INFO	RO	Error information

Table 5: Register Map

Access Types:

- R/W: Read/Write
- RO: Read Only
- R/W1C: Read, Write-1-to-Clear

4.2 CTRL Register (0xF00)

Bits	Name	Access	Reset	Description
31:8	Reserved	RO	0	Reserved for future use
7	TIMEOUT_EN	R/W	0	Enable timeout error detection
6:4	TIMEOUT_VAL	R/W	3'b111	Timeout value (2^{VAL+4} cycles)
3	ERR_INT_EN	R/W	0	Enable error interrupt
2	Reserved	RO	0	Reserved
1	SOFT_RST	R/W	0	Soft reset (self-clearing)
0	ENABLE	R/W	1	Bridge enable

Table 6: CTRL Register

4.3 STATUS Register (0xF04)

Bits	Name	Access	Reset	Description
31:8	Reserved	RO	0	Reserved for future use
7	ERR_INT	R/W1C	0	Error interrupt pending
6	TIMEOUT_ERR	R/W1C	0	Timeout error occurred
5	PSLVERR	R/W1C	0	APB slave error occurred
4	ADDR_ERR	R/W1C	0	Address error occurred
3:2	Reserved	RO	0	Reserved
1	BUSY	RO	0	Bridge busy with transfer
0	READY	RO	1	Bridge ready for transfer

Table 7: STATUS Register

4.4 ERROR_ADDR Register (0xF08)

Bits	Name	Access	Reset	Description
31:0	ERR_ADDR	RO	0	Address that caused error

Table 8: ERROR_ADDR Register

4.5 ERROR_INFO Register (0xF0C)

Bits	Name	Access	Reset	Description
31:16	Reserved	RO	0	Reserved
15:8	ERR_BEAT	RO	0	Beat number where error occurred
7:4	ERR_TYPE	RO	0	Error type (see Table 10)
3	ERR_WRITE	RO	0	Error on write(1) or read(0)
2:0	ERR_SIZE	RO	0	Transfer size when error occurred

Table 9: ERROR_INFO Register

ERR_TYPE	Description
4'h0	No error
4'h1	Address error (invalid address range)
4'h2	Timeout error (APB slave not responding)
4'h3	Protocol error (PSLVERR from slave)
4'h4-F	Reserved

Table 10: Error Type Encoding

5 Protocol Rules

5.1 AHB Protocol Compliance

The bridge shall comply with all AHB-Lite protocol rules:

- AHB-1:** All signal transitions shall occur on the rising edge of HCLK.
- AHB-2:** HREADY_OUT shall be HIGH during IDLE state.
- AHB-3:** When HREADY_OUT is LOW, the master shall hold all control signals stable.
- AHB-4:** HRDATA shall be valid when HREADY_OUT is HIGH and HRESP is OKAY.
- AHB-5:** HRESP shall be OKAY (0) for successful transfers.
- AHB-6:** For ERROR response, HRESP shall be HIGH for two cycles.

5.2 APB Protocol Compliance

The bridge shall comply with all APB protocol rules:

- APB-1:** PSEL shall be asserted for the entire transfer (SETUP and ACCESS phases).
- APB-2:** PENABLE shall be LOW during SETUP phase and HIGH during ACCESS phase.
- APB-3:** PADDR, PWRITE, and PWDATA shall be stable for entire transfer.
- APB-4:** Transfer completes when PENABLE is HIGH and PREADY is HIGH.
- APB-5:** PRDATA shall be sampled when PREADY is HIGH on read transfers.

5.3 Transfer Conversion Rules

- CONV-1:** Each AHB beat shall be converted to one APB transfer.
- CONV-2:** AHB address phase captured when HSEL, HREADY_IN, and HTRANS[1] are HIGH.
- CONV-3:** Bridge shall drive HREADY_OUT LOW during APB SETUP and ACCESS phases.
- CONV-4:** HREADY_OUT shall go HIGH when APB transfer completes (PREADY = 1).
- CONV-5:** PADDR shall equal the captured AHB address.
- CONV-6:** PWRITE shall equal the captured HWRITE.
- CONV-7:** PWDATA shall equal HWDATA (directly forwarded).
- CONV-8:** HRDATA shall equal PRDATA (directly forwarded).

5.4 Burst Handling Rules

- BURST-1:** Bursts shall be converted to sequential APB transfers (one per beat).
- BURST-2:** Bridge shall track burst beat count internally.
- BURST-3:** For INCR bursts, address shall increment by transfer size each beat.
- BURST-4:** For WRAP bursts, address shall wrap at the calculated boundary.
- BURST-5:** Burst shall terminate on HTRANS = IDLE or HTRANS = NONSEQ (new burst).
- BURST-6:** Bridge shall not initiate bursts that cross 1KB boundary (AHB rule).

AHB Burst	Beats	APB Xfers	Address Behavior
SINGLE	1	1	No increment
INCR	Variable	Variable	Increment by HSIZE
INCR4	4	4	Increment by HSIZE
INCR8	8	8	Increment by HSIZE
INCR16	16	16	Increment by HSIZE
WRAP4	4	4	Wrap at 4×HSIZE boundary
WRAP8	8	8	Wrap at 8×HSIZE boundary
WRAP16	16	16	Wrap at 16×HSIZE boundary

Table 11: Burst to APB Conversion

5.5 Wait State Rules

- WAIT-1:** When APB slave asserts `PREADY = 0`, bridge shall hold `HREADY_OUT = 0`.
WAIT-2: Bridge shall not timeout immediately; `TIMEOUT_EN` must be set.
WAIT-3: When timeout occurs, bridge shall abort transfer and set `TIMEOUT_ERR`.
WAIT-4: Minimum APB transfer takes 2 cycles (`SETUP + ACCESS` with `PREADY=1`).
WAIT-5: Maximum wait states determined by `TIMEOUT_VAL` when `TIMEOUT_EN=1`.

6 Reset and Initialization

6.1 Reset Sequence

The bridge supports asynchronous reset assertion and synchronous reset deassertion.

- RST-1:** `HRESETn` may be asserted asynchronously at any time.
RST-2: Upon reset assertion, all outputs shall be driven to safe state within 1 cycle.
RST-3: `HRESETn` deassertion shall be synchronous to rising edge of `HCLK`.
RST-4: Bridge shall be ready for transfers 2 cycles after reset deassertion.
RST-5: Any in-progress transfer shall be aborted on reset.

6.2 Power-On State

After reset, the bridge shall be in the following state:

Signal/Register	Reset Value	Notes
<code>HREADY_OUT</code>	1	Ready for transfers
<code>HRESP</code>	0	OKAY
<code>HRDATA</code>	0	Cleared
<code>PSEL</code>	0	No APB transfer
<code>PENABLE</code>	0	No APB transfer
<code>PADDR</code>	0	Cleared
<code>PWDATA</code>	0	Cleared
<code>CTRL</code>	0x00000001	Enabled, no timeout
<code>STATUS</code>	0x00000001	Ready, no errors
<code>ERROR_ADDR</code>	0x00000000	Cleared
<code>ERROR_INFO</code>	0x00000000	Cleared

Table 12: Reset State

6.3 Soft Reset via Register

- SRST-1:** Writing 1 to CTRL.SOFT_RST shall initiate soft reset.
SRST-2: Soft reset shall abort any in-progress transfer.
SRST-3: Soft reset shall clear STATUS error bits.
SRST-4: Soft reset shall NOT clear CTRL register (except SOFT_RST bit).
SRST-5: SOFT_RST bit shall self-clear after 1 cycle.
SRST-6: Bridge shall be ready 2 cycles after soft reset.

6.4 Recovery from Error

- RCVR-1:** After error, bridge shall return to IDLE state.
RCVR-2: Error information shall be captured in ERROR_ADDR and ERROR_INFO.
RCVR-3: STATUS error bits shall be set and remain set until cleared by software.
RCVR-4: Bridge shall be ready for new transfer after error (no soft reset required).
RCVR-5: Software may clear error bits by writing 1 to the corresponding STATUS bits.

7 Error Handling

7.1 Error Sources

The bridge detects and reports three types of errors:

Error Type	Source	Detection
Address Error	Invalid address range	Bridge internal check
Timeout Error	APB slave not responding	PREADY=0 exceeds timeout
Protocol Error	APB slave error	PSLVERR=1 from slave

Table 13: Error Sources

7.2 Address Error

- AERR-1:** Address error occurs when HADDR is outside valid APB address range.
AERR-2: Valid range is configurable (default: 0x00000000 to 0x0000FFFF).
AERR-3: On address error, bridge shall NOT initiate APB transfer.
AERR-4: Bridge shall respond with HRESP=ERROR on AHB side.
AERR-5: STATUS.ADDR_ERR shall be set.
AERR-6: ERROR_ADDR shall capture the invalid address.
AERR-7: ERROR_INFO.ERR_TYPE shall be set to 4'h1.

7.3 Timeout Error

- TERR-1:** Timeout error occurs when APB slave holds PREADY=0 too long.
TERR-2: Timeout detection requires CTRL.TIMEOUT_EN = 1.
TERR-3: Timeout value = $2^{CTRL.TIMEOUT_VAL+4}$ clock cycles.
TERR-4: On timeout, bridge shall abort APB transfer (PSEL=0, PENABLE=0).
TERR-5: Bridge shall respond with HRESP=ERROR on AHB side.
TERR-6: STATUS.TIMEOUT_ERR shall be set.
TERR-7: ERROR_ADDR shall capture the address of timeout transfer.
TERR-8: ERROR_INFO.ERR_TYPE shall be set to 4'h2.

TIMEOUT_VAL	Timeout Cycles
3'b000	16
3'b001	32
3'b010	64
3'b011	128
3'b100	256
3'b101	512
3'b110	1024
3'b111	2048 (default)

Table 14: Timeout Values

7.4 Protocol Error

- PERR-1:** Protocol error occurs when APB slave asserts PSLVERR=1.
PERR-2: PSLVERR is sampled when PENABLE=1 and PREADY=1.
PERR-3: Bridge shall propagate error to AHB as HRESP=ERROR.
PERR-4: STATUS.PSLVERR shall be set.
PERR-5: ERROR_ADDR shall capture the address of failed transfer.
PERR-6: ERROR_INFO.ERR_TYPE shall be set to 4'h3.
PERR-7: For bursts, error terminates remaining beats.

7.5 Error Reporting

- ERPT-1:** All errors set corresponding bit in STATUS register.
ERPT-2: All errors capture address in ERROR_ADDR register.
ERPT-3: All errors capture details in ERROR_INFO register.
ERPT-4: If CTRL.ERR_INT_EN=1, STATUS.ERR_INT shall be set on any error.
ERPT-5: Error interrupt (if implemented) shall be asserted when ERR_INT=1.
ERPT-6: Multiple errors: only first error captured until cleared.

7.6 Error Recovery

- ERCV-1:** Bridge automatically returns to IDLE after error.
ERCV-2: Software shall read ERROR_ADDR and ERROR_INFO for diagnosis.
ERCV-3: Software shall write 1 to STATUS error bits to clear them.
ERCV-4: No soft reset required for error recovery.
ERCV-5: Bridge accepts new transfer immediately after error completion.

8 Timing Diagrams

8.1 Single Write Transfer

A single write transfer converts one AHB write to one APB write.

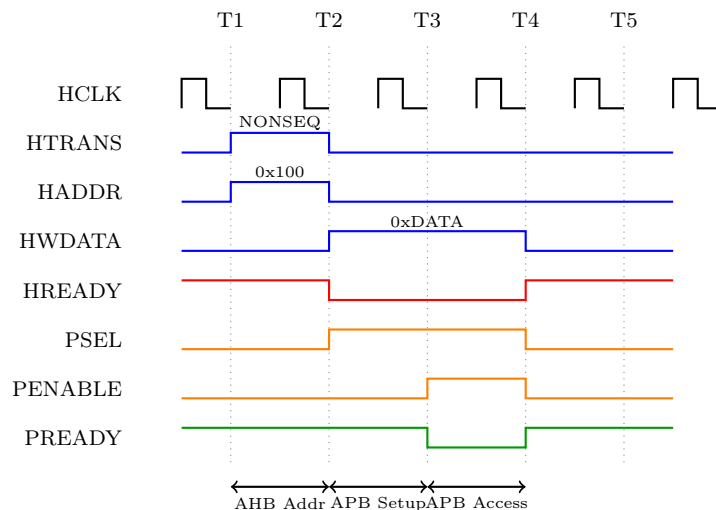


Figure 2: Single Write Transfer Timing

8.2 Single Read Transfer

A single read transfer returns data from APB slave to AHB master.

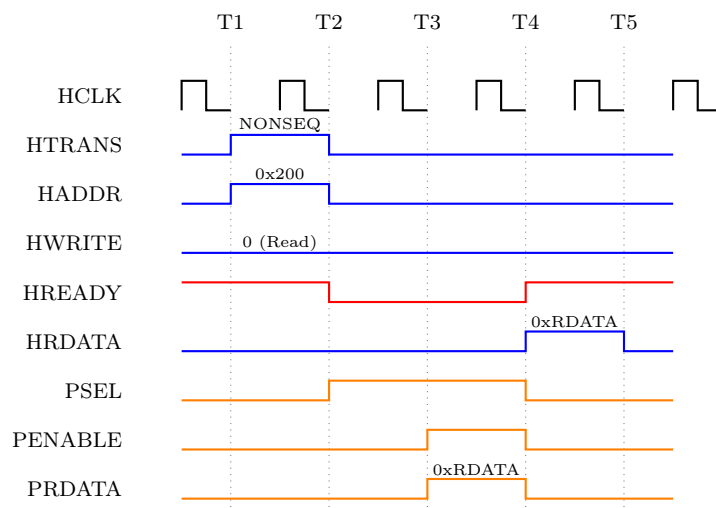


Figure 3: Single Read Transfer Timing

8.3 INCR4 Burst Write

A 4-beat incrementing burst converts to 4 sequential APB writes.

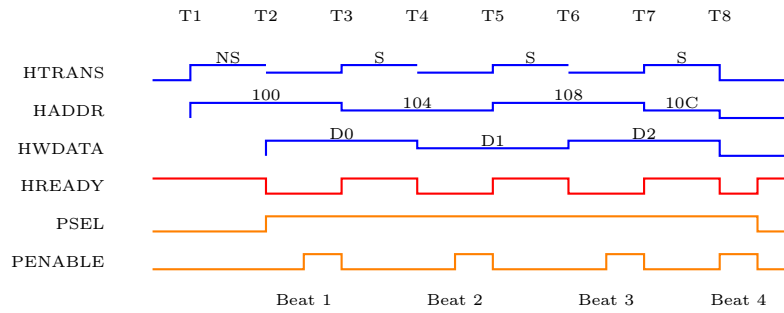


Figure 4: INCR4 Burst Write Timing

8.4 Transfer with APB Wait States

When APB slave asserts PREADY=0, bridge inserts wait states.

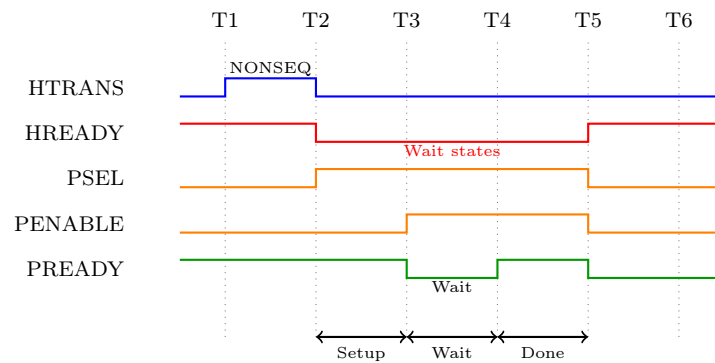


Figure 5: Transfer with APB Wait States

8.5 Error Response Timing

When error occurs, bridge responds with HRESP=1 for two cycles (AHB protocol).

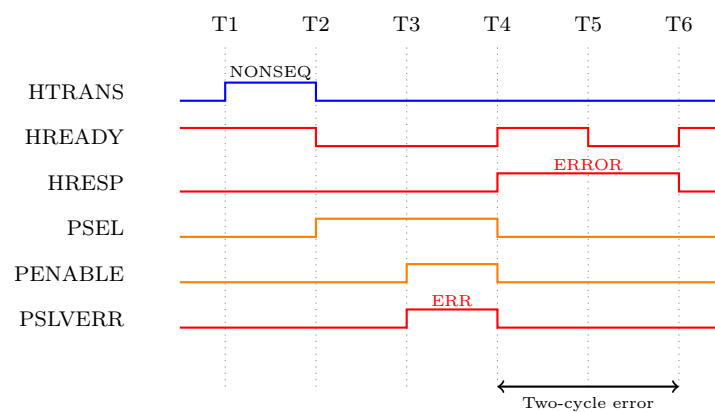


Figure 6: Error Response Timing

A AHB Protocol Summary

A.1 Transfer Types

HTRANS	Type	Description
2'b00	IDLE	No transfer required
2'b01	BUSY	Insert wait cycle in burst
2'b10	NONSEQ	First transfer or single
2'b11	SEQ	Remaining transfers in burst

Table 15: HTRANS Encoding

A.2 Address Increment

For incrementing bursts, address increments by transfer size:

HSIZE	Size	Increment
3'b000	Byte	+1
3'b001	Halfword	+2
3'b010	Word	+4

Table 16: Address Increment by HSIZE

B APB Protocol Summary

B.1 State Machine

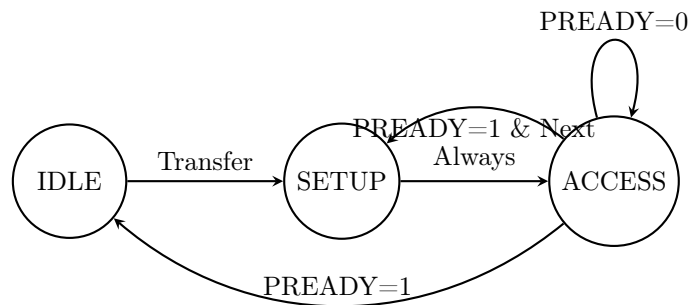


Figure 7: APB State Machine

B.2 Signal Timing

Signal	IDLE	SETUP	ACCESS
PSEL	0	1	1
PENABLE	0	0	1
PADDR	X	Valid	Valid
PWRITE	X	Valid	Valid
PWDATA	X	Valid	Valid

Table 17: APB Signal States

C Revision History

Version	Date	Author	Changes
1.0	Feb 2026	Vikash Kumar	Initial release

Table 18: Document Revision History